PROMPT — "Copyright Match Tool": YouTube-link flow + YT Studio uploader + model/window rules

Tum is repo (Next.js + Playwright, `Copyright-matchtool-25-main`) ke senior engineer ho. Pehle `AGENTS.md` padho (is Next.js version me breaking changes hain — code likhne se pehle `node_modules/next/dist/docs/` check karo). Neeche ka poora kaam karo. Jo cheez already kaam kar rahi hai use mat todo: Gemini prompt, 24 fps, HISSA 1/HISSA 2 parsing, offset calibration, results/report UI — ye sab same rahega.

---

0. Background (kyun badal rahe hain)

App ka Parallel Window Scan abhi aise chalta hai:

1. Movie ko windows (default 5 min) me ffmpeg se kaata jata hai (`lib/studio-window-clips.ts`, `extractWindowClip`) -> clips `data/media/<scanId>/clips/*.mp4`.
2. Har window ke liye `lib/studio-automation.ts` -> `attachVideoAndConfigureRange()` Google AI Studio (`aistudio.google.com/prompts/new_chat`) me video attach karta hai: Insert menu -> Drive picker, ya `uploadLocalFileToStudio()` se local file (390 MB tak, 5 min wait).
3. Start/End time + FPS=24 set karke Run, output parse.

Problem: bade local/Drive uploads par AI Studio "internal error" deta hai. **Solution mil gaya:** AI Studio YouTube video link ko bina error ke leta hai. Movie clips YouTube par upload karne par YouTube unhe copyright se *block* kar deta hai, lekin block hone ke baad bhi uska **share link kaam karta hai**, aur AI Studio us link se video dekh leta hai. Isliye naya flow:

```
movie window clip (ffmpeg)  ->  YouTube Studio par upload (same channel, hamesha)  ->  share link lo
  ->  AI Studio me Insert -> YouTube Video -> link paste -> 24 fps -> baaki sab same
  ->  scan khatam hone par channel ke saare (blocked) videos delete
```

Short video ke liye: short already YouTube par hota hai, to short ka **YouTube link** dene ka option do (upload ya blocking ki zarurat nahi).

Note: user jab "yt studio me upload button -> youtube video -> upar link -> 24 fps" bolta hai to wo **AI Studio ka attach step** hai (Insert/Upload button -> "YouTube Video" -> top input me link). Aur "YT Studio cookies / yt panel" matlab **YouTube Studio** (studio.youtube.com) ka alag login/session. Dono ko confuse mat karna.

---

1. AI Studio attach: YouTube link support

File: `lib/studio-automation.ts`, `attachVideoAndConfigureRange()` (~line 4052) aur uska caller (~line 5950-6070).

- Params me `youtubeUrl?: string | null` add karo. Priority: **youtubeUrl > driveFileName > localFilePath**. (Drive aur local ko fallback ke taur par rehne do, par default ab YouTube.)
- Flow: Insert/Upload button (`STUDIO_SELECTORS.insertMediaButton`) -> `insertMenuYoutubeOption` ("YouTube Video") -> modal ke **sabse upar wale input** me link paste (`youtubeModalInput`) -> `youtubeModalConfirm` -> chip aane ka wait. Selectors `lib/studio-selectors.ts` (~line 219-312) me already hain (`ms-youtube-chunk`, `.chunk-chip:has-text("youtube")`); agar AI Studio UI badla ho to selectors update karo aur fallback selectors rakho.
- Chip load hone ke baad **wahi existing logic** chalao: Video options gear -> Start/End time -> **FPS = 24** -> read-back verify -> Save (jaisa abhi local/Drive me hota hai). Chip count / token-count success signals same rahenge; `ms-youtube-chunk` ko `videoMediaChipsCss` me count karo.
- YouTube attach me bada upload nahi hota, isliye local-upload wala 300 s timeout aur RAM-gate (`waitUntilRamSafeForAttach`) sirf local fallback par lagao. YouTube attach ke liye timeout ~90 s, aur retry (3 baar, beech me 20-30 s) kyunki naya upload hua video kabhi kabhi turant processed nahi hota.
- Agar attach fail ho: wahi `dumpChipAttachDiagnostics` + log. Internal-error wale refresh/restart logic (`MAX_INTERNAL_ERROR_REFRESH`) same rahe.

2. Short video: upload RAHEGA + niche YouTube link option

Short upload hatana nahi hai. Short video ka local upload zaruri rahega, kyunki Minute Finder windows aur Backup Minute Finder (`lib/minute-finder-dispatch.ts`, `lib/batch-verifier-service.ts`, `data/gemini-*minute-finder-prompt.md`) isi uploaded short (mute) ko use karte hain. Is flow ko bilkul mat chhedo.

Naya option sirf **AI Studio window scan ke attach step** ke liye hai:

- `components/cmt/upload-panel.tsx` me short upload field ke **neeche** "Short YouTube link" input + do buttons: **Save link** aur **Cancel** (Cancel = input clear + saved link hata do, scan wapas local-short attach par).
- Link `isValidYoutubeUrl()` (`lib/yt-dlp.ts`) se validate karo; invalid par inline error, save mat karo.
- Save par scan record me `shortYoutubeUrl` store karo (`lib/types.ts`, `lib/scan-store.ts`, `app/api/scans/[id]/...` ke naye route/PATCH se). Saved link ka badge dikhao (e.g. "Short link saved") + Remove/Cancel.
- `shortYoutubeUrl` ho to AI Studio me short **seedha link se** attach hoga (section 1 ka flow, `youtubeUrl`), local/Drive attach skip. Short ke start/end (`studioShortPartMinutes`) aur 24 fps wahi AI Studio video settings se.
- `shortYoutubeUrl` na ho to purana flow (local short attach, `ensureStudioShortReady`) jaisa tha waisa.
- Short ka link sirf window scan attach me use hoga; Minute Finder / Backup Minute Finder hamesha uploaded local short par chalenge.
- Parallel manager (`lib/studio-parallel-manager.ts`, `shortDriveFileName` wali jagah) me `shortYoutubeUrl` pass-through karo.

---

3. NAYA module: YouTube Studio uploader (`lib/yt-studio-automation.ts`)

Playwright se `studio.youtube.com` control karne wala alag class, AI Studio wale class (`StudioAutomation`) jaisa pattern: **alag persistent profile dir** (AI Studio profile se alag), headless Chromium, `assistMode`, snapshot/screenshot, `logStep`.

Zaruri behaviour:

1. **Session = YT Studio cookies.** User settings me JSON paste karega (Cookie-Editor array ya Playwright storageState). `lib/studio-settings.ts` ka `parseCookieInputToPlaywright`, `encryptSecret/decryptSecret` reuse karo. Naye fields `StudioUserConfig` + `PublicStudioSettings` me: `encryptedYtCookies`, `hasYtCookies`, `ytCookieCount`, `ytChannelId` / `ytChannelHandle` (locked channel), `ytLastLoginStatus` (`logged_in|logged_out|untested`), `ytLastLoginTestedAt`, `ytVisibility` (default `unlisted`), `ytAutoDeleteAtEnd` (default `true`), `ytUploadConcurrency` (default 3). Cookies kabhi client ko return mat karo (jaise Google cookies ke saath hota hai).
2. **Hamesha wahi channel.** Test Login par channel id/handle detect karke save karo (pehli baar lock). Har upload aur har delete se pehle current channel ko locked channel se

compare karo; mismatch ho to **ruk jao**, error log karo, kuch upload/delete mat karo.

3. **Upload:** studio.youtube.com -> Create/Upload videos -> file chooser me **local clip file direct** `setInputFiles` se (Drive nahi, jaise AI Studio me local upload hota hai) -> title (e.g. `cmt-<scanId>-w<windowNo>`), "Not made for kids", visibility `ytVisibility` (Unlisted/Public; **Unlisted hi default/forced rakho** (Private mat rakho — AI Studio private link nahi dekh sakta; Public bhi nahi)) -> Next/Next/Next -> Save/Publish. Copyright "Checks" step me claim/block aaye to bhi **aage badho aur Done/Save karo** — video block ho jayega par Unlisted link working rehta hai (user ne khud test kiya hai). Har clip isi tarah upload karo. Dialog me aane wala **share link / video id** capture karo (`https://youtu.be/<id>`).
4. Upload ke baad `ytUrl`, `ytVideoId`, `uploadedAt`, `clipSize/mtime` ko clip ke sidecar JSON (`getWindowClipSidecarPath`) aur scan state me save karo. Same clip dobara aaye (resume/retry) to **reuse** karo, dobara upload mat karo (jab tak clip file change na hui ho).
5. Errors: login expire (redirect to accounts.google.com) -> status `logged_out` + clear log; daily upload limit / "verify your account" -> pause + clear message; upload stuck -> timeout (size ke hisaab se) + retry 2 baar. Har step ka log `StudioStepLog` format me.
6. Concurrency: `ytUploadConcurrency` (default 3) ka global upload gate (existing `GlobalStudioUploadGate` ka pattern, alag instance).

4. Pipeline wiring (parallel manager)

`lib/studio-parallel-manager.ts`: har window job ke liye order:

1. window clip extract (existing),
2. "**uploading to YouTube**" stage: `yt-studio-automation` se upload -> `ytUrl`,
3. AI Studio job ko `movieYoutubeUrl = ytUrl` do -> `attachVideoAndConfigureRange({ youtubeUrl })`,
4. baaki run/parse/validate **bilkul same** (offset mode, validator, retry logic).

Clip upload aur AI Studio runs ko **pipeline** karo: jab window N AI Studio me chal rahi ho tab window N+1..N+k ke clips background me YT par upload hote rahein (limit `ytUploadConcurrency`). UI me har window ka naya state dikhao: `clip ready -> uploading to YT -> link ready -> attaching -> running -> done`.

`studioAttachMovieFromDrive` aur Drive-movie path ko deprecated rakho (fallback only, default OFF). `studioUploadGateConcurrency` ab sirf local fallback ke liye.

Auto-start (manual step nahi)

Ye poora YouTube flow **Start Parallel Window Scan** (`start_parallel_scan` in `app/api/studio/control/route.ts` , `lib/studio-parallel-manager.ts`) dabate hi **apne aap** chalu hona chahiye: YT session check (locked channel) -> window clips extract -> YT upload -> link -> AI Studio attach. Alag se "upload to YT" button dabane ki zarurat na ho. Start par agar YT cookies/login missing ya expired ho to scan shuru hone se pehle clear error dikhao ("YT Studio login required") aur start roko.

---

5. End-of-scan cleanup (channel saaf karna)

- Jab scan **poora khatam** ho (sab windows done/failed-final/stopped — koi retry pending nahi) tab `ytAutoDeleteAtEnd` ON ho to: YT Studio -> Content -> videos list -> **Select all (checkbox) -> Delete forever** -> confirm checkbox -> delete. Pagination ho to sab pages tak repeat jab tak list khali na ho.
- Is channel par koi aur long video nahi hai, isliye "select all delete" hi sahi hai — **lekin** delete se pehle locked channel verify karna zaruri hai (section 3.2). Mismatch par kabhi delete mat karo.
- Cleanup scan stop/fail par bhi chale (finally-block), par kisi *dusre running scan* ke clips ke beech me nahi: agar koi aur scan active hai to cleanup us scan ke khatam hone tak defer karo.
- Manual button bhi do: "Delete all videos from channel now" (confirm dialog ke saath). Result log me: kitne videos delete hue.
- Delete hone ke baad scan me stored `ytUrl` ko `deleted` mark karo, taaki retry par dobara upload ho.

6. Model & window rules (SIMPLIFY)

Files: `lib/studio-model-state.ts` , `lib/studio-settings.ts` , `lib/studio-parallel-manager.ts` , `lib/studio-automation.ts` , `components/cmt/studio-settings-section.tsx` , `studio-parallel-grid.tsx` .

1. **"1 window = 1 model" limit hatao.** `MAX_CONCURRENT_MODEL_WINDOWS` (= 3) aur lease system (`updateExclusiveStudioModelLease` , `releaseExclusiveStudioModel` , `orderChainByUnleasedFirst` , `orderFallbacksKeepingTargetFirst` , `getActiveLeasedModelsByOtherHolders`) remove karo. Kai windows **same model** use kar sakti hain.
2. **10 windows ek saath.** Concurrency cap ko 10 karo (`studioConcurrency` clamp abhi max 20 / manager me `Math.min(MAX_CONCURRENT_MODEL_WINDOWS, …)` — dono jagah 10).

Default ab 10 rakho (agar RAM warning ho to UI me dikhao, par block mat karo). `maxTotalPoolSlots` aur RAM-guard logic check karo ki 10 Chromium windows par sahi kaam kare; stagger (`studioRunStaggerSec`) rahe.

3. **Settings wala model hi sab chalayein.** Sab windows `studioModelName` (Argon / Gemini 3.8 Flash / Gemini 3.6 Flash, thinking level, max output tokens rules same) use karein.
4. **Model sirf quota-over par badlo.** Jab AI Studio quota/daily-limit exceeded message aaye tabhi `studioExhaustedModels` me mark karo (reset time `QUOTA_RESET_TZ`) aur `buildStudioModelChain()` ka next model chuno. Ye switch **shared state** se ho (ek model exhausted -> sab windows next model par), har window alag alag thrash na kare.
5. **Internal error par model switch NAHI.** `studio-automation.ts` me jahan "internal error -> switch model" hota hai (~7227-7270) wo hatao; internal error par sirf existing hard-refresh -> new page -> browser restart -> naya chat retry chale, same model par. (Naye YouTube flow me internal error waise bhi kam hona chahiye.)

7. UI: YouTube Studio panel (Gemini Studio panel jaisa)

`components/cmt/studio-live-panel.tsx` aur `studio-settings-section.tsx` ko dekh kar **same design/buttons** ka naya panel `components/cmt/yt-studio-panel.tsx` banao:

- Live browser screen (screenshot stream, click/type/key relay) — **Login Assist** toggle (interactive), **Test Login**, **Enter/Save session** button (jo abhi Gemini panel me hai: `assist_toggle`, `test_login`, `save_session`), Pause/Resume/Stop.
- Screen ke **neeche logs** (events list, jaise Gemini panel me), aur live view par click karne se wahi action browser me jaye (`click`, `type`, `key`, `navigate`).
- Cookies textarea: "YT Studio cookies" paste -> Save Encrypted / Replace / Clear; status badge (cookies count, logged in/out, channel handle).
- Settings: visibility, auto-delete toggle, upload concurrency, "Delete all videos now", "Test upload" (chhota test clip upload karke link dikhao).
- Panel bhi **direct upload** dikhaye: local clip -> YT Studio, bina Drive ke (jaise Gemini Studio panel local upload karta hai).

API: `app/api/studio/control/route.ts` me `yt_*` actions (`yt_save_cookies`, `yt_test_login`, `yt_assist_toggle`, `yt_click`, `yt_type`, `yt_key`, `yt_navigate`, `yt_upload_test`, `yt_cleanup_now`) ya alag `app/api/yt/*` routes; live snapshot `app/api/studio/stream/route.ts` me YT snapshot bhi add karo. Auth/session checks existing routes jaise (`proxy.ts`, `lib/session.ts`).

8. Yaad rakhne wali baatein (non-negotiable)

- Movie clips YouTube par block ho jaate hain — **ye normal hai**. Block hone ko error mat maano; link working hota hai.
- Hamesha **same YouTube channel**; channel mismatch par kuch mat karo.
- Kaam ke baad (scan end) channel ke saare videos delete — warna channel par blocked clips jama honge.
- FPS hamesha **24**; short aur movie dono ke start/end AI Studio video settings se hi set honge.
- Gemini prompt aur result parsing **bina badlaav** ke.
- Cookies/secrets sirf encrypted store honge, logs me kabhi print nahi honge.

9. Delivery

1. Pehle chhota plan likho (kaun si files badlengi), phir implement karo.
2. `npm run lint` + `tsc --noEmit` clean hona chahiye; existing scripts (`scripts/test-studio-*.mjs`) fail na hon.
3. Naye unit-style tests: YouTube URL validation, cookie parse/encrypt round-trip, clip-sidecar reuse (same clip -> dobara upload nahi), model switch sirf quota par, concurrency clamp = 10.
4. Manual test checklist final reply me do: (a) YT cookies save + Test Login, (b) test clip upload -> link, (c) AI Studio me link attach + 24 fps verify, (d) 10 windows parallel same model, (e) quota message par hi model switch, (f) scan end par channel cleanup.
5. Jo cheez UI selectors ki wajah se live verify nahi kar sake (YouTube/AI Studio UI badal sakta hai) usko clearly "unverified" bolo aur fallback selectors + diagnostics screenshot (`data/studio-attach-fail-*.png` jaisa) rakho.